FairCKD: Fairness Audit Report

Investigating Bias and Fairness in Machine Learning Models for Automated CKD Diagnosis
M.Tech Dissertation Phase-II · Jimil Jingar · PIET, Parul University

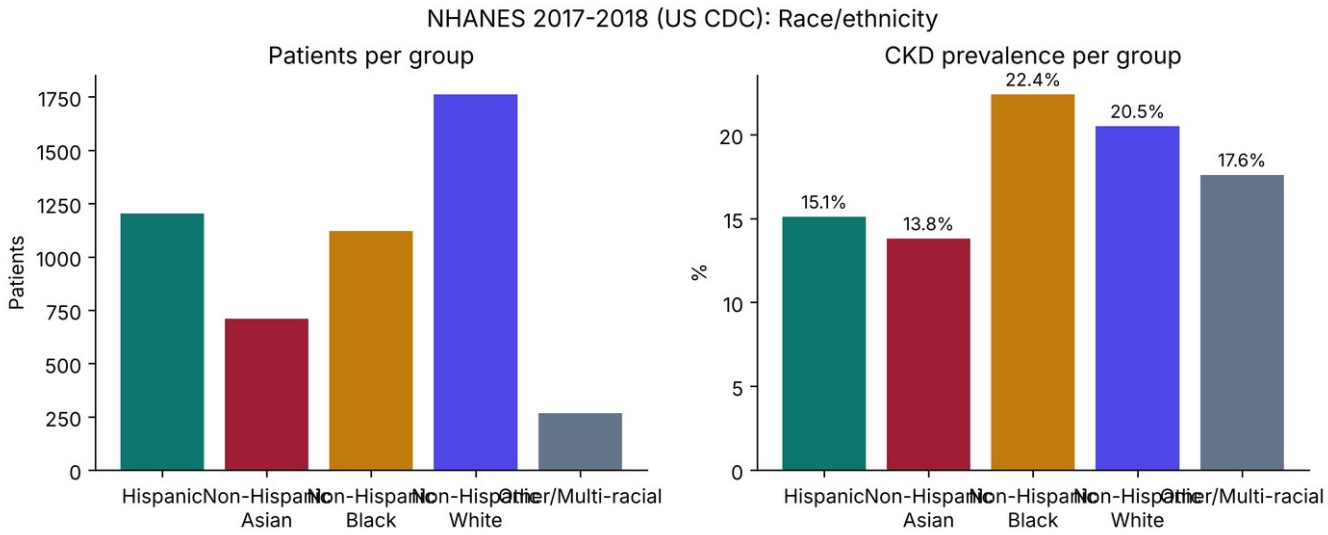
	Value
Dataset	NHANES 2017-2018 (US CDC)
Feature set	Routine blood tests (17 features, no kidney-function tests)
Sensitive attribute	Race/ethnicity
Patients	5063
Features	17
Groups	Hispanic, Non-Hispanic Asian, Non-Hispanic Black, Non-Hispanic White, Other/Multi-racial
Decision threshold	training-set CKD prevalence
CV protocol	stratified 5-fold, 5 repeats (baseline), 5 repeats (mitigation)

Key findings

• Best baseline model: **Random Forest** (accuracy 0.740, balanced accuracy 0.738, AUC 0.805).
• Bias detection: in **7 of 7** models, CKD is detected least often in the **Non-Hispanic Asian** group (Equal Opportunity gap 0.165–0.259).
• Even a perfect classifier has DP diff = 0.086 and DI = 0.617 here, because CKD prevalence differs between groups; error-rate metrics (Equal Opportunity, Equalized Odds) are the clinically meaningful fairness criteria.
• Logistic Regression: best mitigation (balanced accuracy loss ≤ 3 points) = **Exp. Gradient**, Equalized Odds gap 0.204 → 0.092, balanced accuracy 0.735 → 0.720.
• XGBoost: best mitigation (balanced accuracy loss ≤ 3 points) = **Exp. Gradient**, Equalized Odds gap 0.259 → 0.222, balanced accuracy 0.736 → 0.729.
• **FairCKD-Select** (proposed, nested CV): Equalized Odds gap 0.220 → **0.128** compared with accuracy-driven model selection, balanced accuracy 0.733 → 0.719.
• Statistical significance: 0 of 8 mitigation results reduce the Equalized Odds gap significantly (paired bootstrap, $p < 0.05$).
• External validation (NHANES 2015–16): baseline balanced accuracy 0.740–0.749, Equalized Odds gap 0.159–0.199; best mitigated gap 0.086.
• Label bias: the race-based 2009 eGFR equation changes CKD prevalence for Non-Hispanic Black adults from 22.4% to 19.3% (35 patients relabelled), i.e. the ground-truth label itself can carry bias.

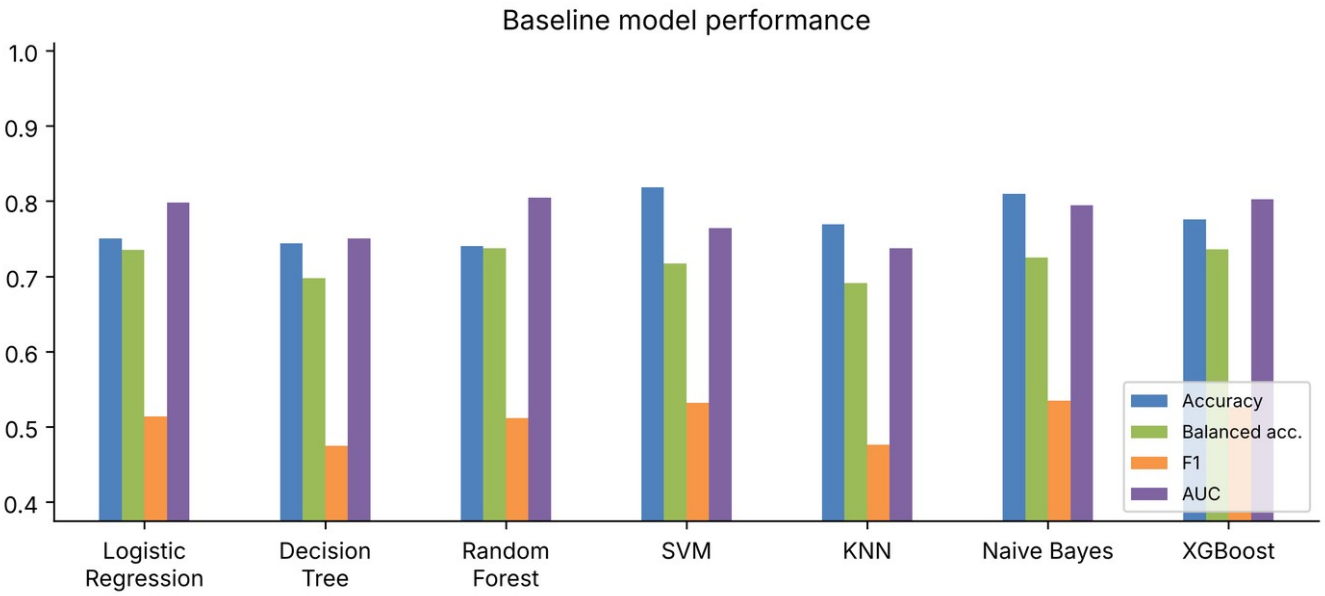
1. Dataset and groups

Group	Patients	Share (%)	CKD cases	CKD prevalence (%)	Missing values / patient
Hispanic	1204	23.8	182	15.1	0.190
Non-Hispanic Asian	709	14	98	13.8	0.170
Non-Hispanic Black	1121	22.1	251	22.4	0.140
Non-Hispanic White	1762	34.8	361	20.5	0.130
Other/Multi-racial	267	5.3	47	17.6	0.100
All	5063	100	939	18.5	0.150


NHANES 2017-2018 (US CDC): Race/ethnicity
Patients per group
Patients
1750
1500
1250
1000
750
500
250
0
Hispanic
Non-Hispanic Asian
Non-Hispanic Black
Non-Hispanic White
Other/Multi-racial
CKD prevalence per group
%
20
15
10
5
0
15.1%
13.8%
22.4%
20.5%
17.6%
Hispanic
Non-Hispanic Asian
Non-Hispanic Black
Non-Hispanic White
Other/Multi-racial

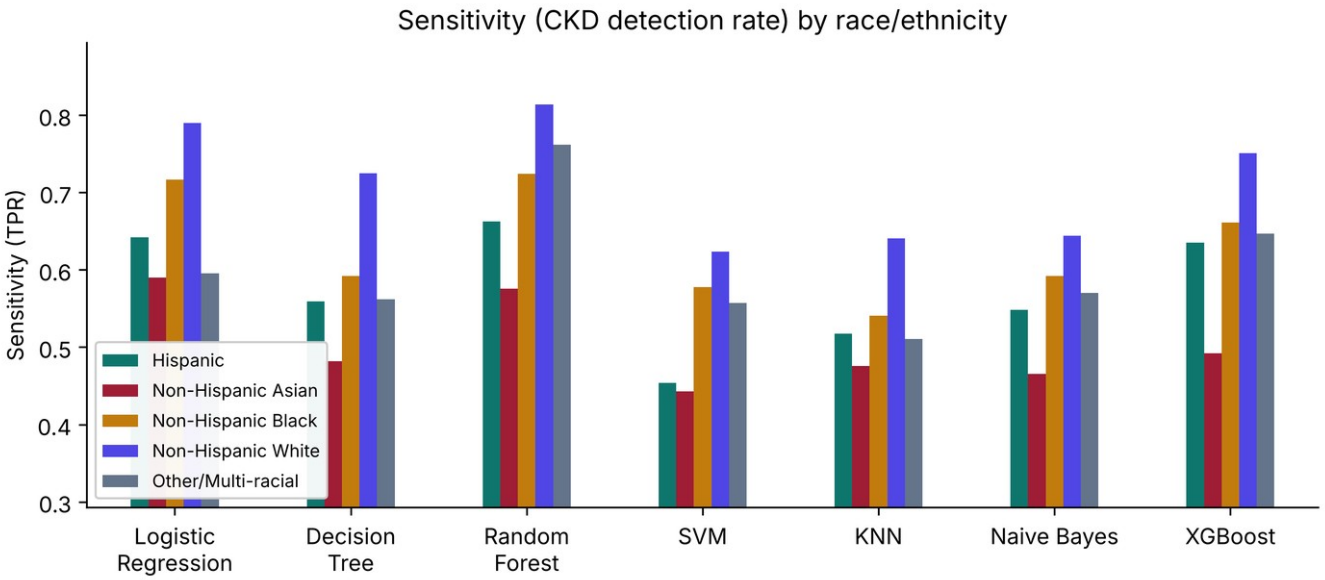
2. Baseline model performance

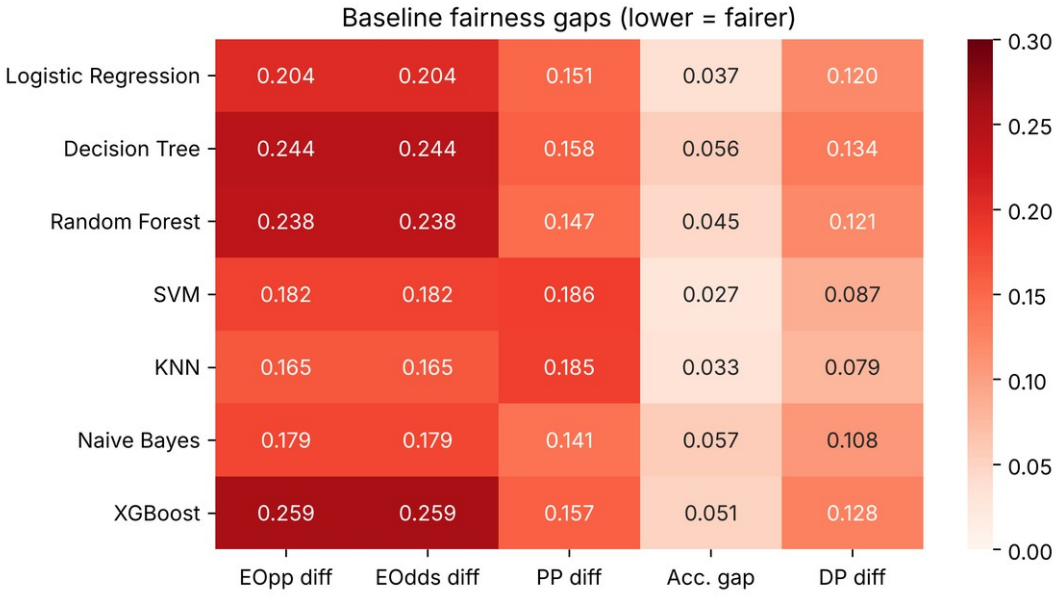
	Accuracy	Precision	Recall	F1	AUC	Balanced acc.
Logistic Regression	0.750 ± 0.001	0.402 ± 0.002	0.711 ± 0.002	0.514 ± 0.002	0.798 ± 0.001	0.735 ± 0.001
Decision Tree	0.744 ± 0.009	0.383 ± 0.012	0.624 ± 0.013	0.475 ± 0.010	0.750 ± 0.004	0.698 ± 0.007
Random Forest	0.740 ± 0.001	0.393 ± 0.001	0.733 ± 0.008	0.512 ± 0.003	0.805 ± 0.002	0.738 ± 0.003
SVM	0.819 ± 0.002	0.510 ± 0.005	0.556 ± 0.007	0.532 ± 0.005	0.764 ± 0.004	0.717 ± 0.003
KNN	0.769 ± 0.002	0.411 ± 0.004	0.566 ± 0.012	0.476 ± 0.006	0.737 ± 0.004	0.691 ± 0.005
Naive Bayes	0.810 ± 0.002	0.490 ± 0.005	0.589 ± 0.003	0.535 ± 0.003	0.795 ± 0.001	0.725 ± 0.002
XGBoost	0.776 ± 0.004	0.433 ± 0.006	0.672 ± 0.005	0.526 ± 0.005	0.802 ± 0.004	0.736 ± 0.003



3. Bias detection and fairness evaluation

	DP diff	DI ratio	EOpp diff	EOdds diff	PP diff	Acc. gap
Logistic Regression	0.120 ± 0.005	0.680 ± 0.010	0.204 ± 0.012	0.204 ± 0.012	0.151 ± 0.012	0.037 ± 0.005
Decision Tree	0.134 ± 0.015	0.638 ± 0.037	0.244 ± 0.034	0.244 ± 0.034	0.158 ± 0.019	0.056 ± 0.003
Random Forest	0.121 ± 0.009	0.696 ± 0.021	0.238 ± 0.015	0.238 ± 0.015	0.147 ± 0.012	0.045 ± 0.011
SVM	0.087 ± 0.004	0.637 ± 0.016	0.182 ± 0.005	0.182 ± 0.005	0.186 ± 0.022	0.027 ± 0.007
KNN	0.079 ± 0.008	0.732 ± 0.026	0.165 ± 0.030	0.165 ± 0.030	0.185 ± 0.012	0.033 ± 0.009
Naive Bayes	0.108 ± 0.006	0.597 ± 0.017	0.179 ± 0.007	0.179 ± 0.007	0.141 ± 0.013	0.057 ± 0.003
XGBoost	0.128 ± 0.006	0.622 ± 0.019	0.259 ± 0.027	0.259 ± 0.027	0.157 ± 0.008	0.051 ± 0.007
Perfect classifier (reference)	0.086	0.617	0.000	0.000	0.000	0.000


Sensitivity (CKD detection rate) by race/ethnicity
Sensitivity (TPR)
0.8
0.7
0.6
0.5
0.4
0.3
Hispanic
Non-Hispanic Asian
Non-Hispanic Black
Non-Hispanic White
Other/Multi-racial
Logistic Regression
Decision Tree
Random Forest
SVM
KNN
Naive Bayes
XGBoost


Baseline fairness gaps (lower = fairer)
Logistic Regression 0.204 0.204 0.151 0.037 0.120
Decision Tree 0.244 0.244 0.158 0.056 0.134
Random Forest 0.238 0.238 0.147 0.045 0.121
SVM 0.182 0.182 0.186 0.027 0.087
KNN 0.165 0.165 0.185 0.033 0.079
Naive Bayes 0.179 0.179 0.141 0.057 0.108
XGBoost 0.259 0.259 0.157 0.051 0.128
EOpp diff
EOdds diff
PP diff
Acc. gap
DP diff
0.30
0.25
0.20
0.15
0.10
0.05
0.00

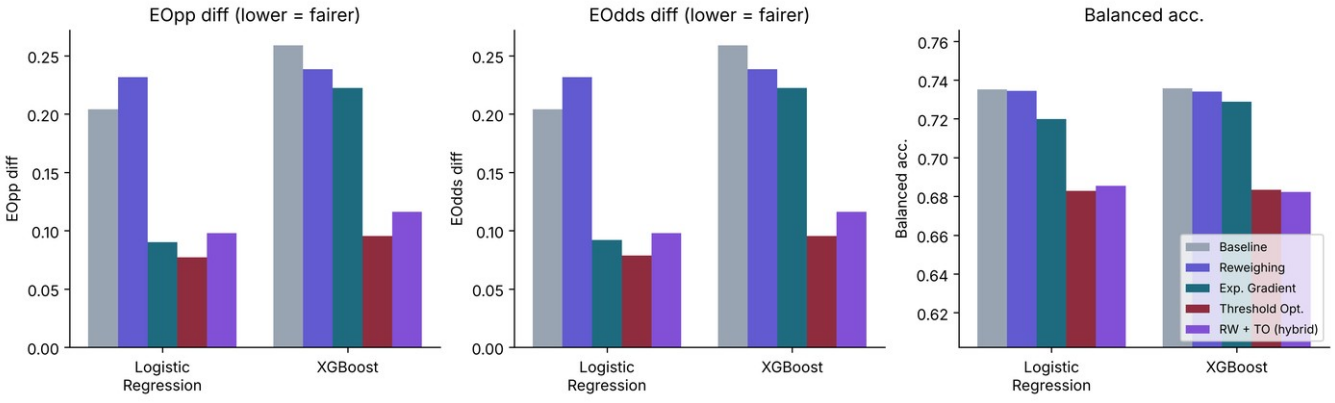
4. Bias mitigation

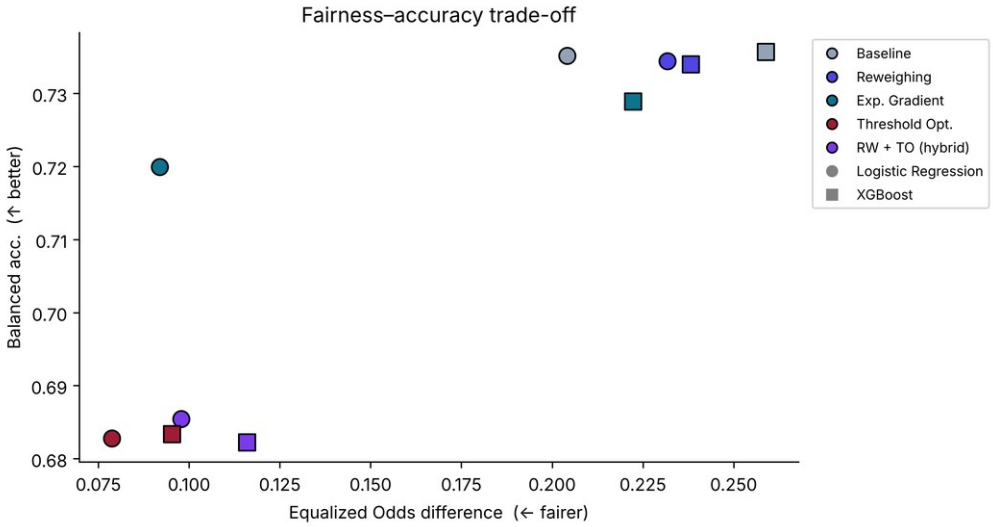
a) Performance (mean ± SD over CV repeats)

Model	Method	Accuracy	Balanced acc.	Recall	F1
Logistic Regression	Baseline	0.750 ± 0.001	0.735 ± 0.001	0.711 ± 0.002	0.514 ± 0.002
Logistic Regression	Reweighing	0.750 ± 0.001	0.734 ± 0.002	0.709 ± 0.004	0.513 ± 0.003
Logistic Regression	Exp. Gradient	0.745 ± 0.003	0.720 ± 0.005	0.681 ± 0.009	0.497 ± 0.006
Logistic Regression	Threshold Opt.	0.748 ± 0.021	0.683 ± 0.005	0.579 ± 0.033	0.460 ± 0.010
Logistic Regression	RW + TO (hybrid)	0.751 ± 0.015	0.685 ± 0.006	0.582 ± 0.016	0.464 ± 0.011
XGBoost	Baseline	0.776 ± 0.004	0.736 ± 0.003	0.672 ± 0.005	0.526 ± 0.005
XGBoost	Reweighing	0.773 ± 0.003	0.734 ± 0.002	0.671 ± 0.002	0.524 ± 0.003
XGBoost	Exp. Gradient	0.791 ± 0.003	0.729 ± 0.005	0.630 ± 0.008	0.528 ± 0.007
XGBoost	Threshold Opt.	0.739 ± 0.025	0.683 ± 0.010	0.595 ± 0.049	0.458 ± 0.013
XGBoost	RW + TO (hybrid)	0.736 ± 0.013	0.682 ± 0.009	0.597 ± 0.025	0.456 ± 0.011

b) Fairness gaps (lower = fairer; DI ratio: 1 = fair)

Model	Method	EOpp diff	EOdds diff	PP diff	DP diff	DI ratio	Acc. gap
Logistic Regression	Baseline	0.204 ± 0.012	0.204 ± 0.012	0.151 ± 0.012	0.120 ± 0.005	0.680 ± 0.010	0.037 ± 0.005
Logistic Regression	Reweighing	0.232 ± 0.024	0.232 ± 0.024	0.151 ± 0.005	0.126 ± 0.009	0.662 ± 0.022	0.032 ± 0.007
Logistic Regression	Exp. Gradient	0.090 ± 0.038	0.092 ± 0.037	0.161 ± 0.011	0.078 ± 0.009	0.779 ± 0.025	0.049 ± 0.011
Logistic Regression	Threshold Opt.	0.077 ± 0.027	0.079 ± 0.025	0.139 ± 0.013	0.037 ± 0.016	0.874 ± 0.058	0.034 ± 0.017
Logistic Regression	RW + TO (hybrid)	0.098 ± 0.023	0.098 ± 0.023	0.155 ± 0.017	0.036 ± 0.009	0.878 ± 0.035	0.038 ± 0.020
XGBoost	Baseline	0.259 ± 0.027	0.259 ± 0.027	0.157 ± 0.008	0.128 ± 0.006	0.622 ± 0.019	0.051 ± 0.007
XGBoost	Reweighing	0.238 ± 0.018	0.238 ± 0.018	0.158 ± 0.016	0.121 ± 0.009	0.644 ± 0.026	0.049 ± 0.010
XGBoost	Exp. Gradient	0.222 ± 0.014	0.222 ± 0.014	0.175 ± 0.014	0.103 ± 0.009	0.656 ± 0.027	0.040 ± 0.014
XGBoost	Threshold Opt.	0.095 ± 0.047	0.095 ± 0.047	0.172 ± 0.015	0.043 ± 0.021	0.862 ± 0.064	0.038 ± 0.007
XGBoost	RW + TO (hybrid)	0.116 ± 0.071	0.116 ± 0.071	0.150 ± 0.019	0.051 ± 0.011	0.841 ± 0.037	0.036 ± 0.009





c) Statistical significance (paired bootstrap vs. baseline, 1000 resamples; Δ = method – baseline)

Model	Method	Δ EOdds gap	95% CI (EOdds)	p (EOdds)	Δ Balanced acc.	95% CI (Bal. acc.)	p (Bal. acc.)
Logistic Regression	Reweighing	0.011	[-0.025, 0.057]	0.644	-0.004	[-0.009, 0.001]	0.158
Logistic Regression	Exp. Gradient	-0.085	[-0.180, 0.024]	0.120	-0.012	[-0.025, 0.001]	0.074
Logistic Regression	Threshold Opt.	-0.098	[-0.206, 0.034]	0.126	-0.052	[-0.066, -0.039]	0.000
Logistic Regression	RW + TO (hybrid)	-0.070	[-0.227, 0.093]	0.372	-0.058	[-0.071, -0.044]	0.000
XGBoost	Reweighing	-0.025	[-0.071, 0.022]	0.254	0.000	[-0.008, 0.008]	0.986
XGBoost	Exp. Gradient	-0.035	[-0.098, 0.028]	0.268	-0.008	[-0.018, 0.001]	0.066
XGBoost	Threshold Opt.	-0.067	[-0.210, 0.073]	0.344	-0.053	[-0.067, -0.040]	0.000
XGBoost	RW + TO (hybrid)	-0.042	[-0.194, 0.114]	0.580	-0.059	[-0.073, -0.044]	0.000

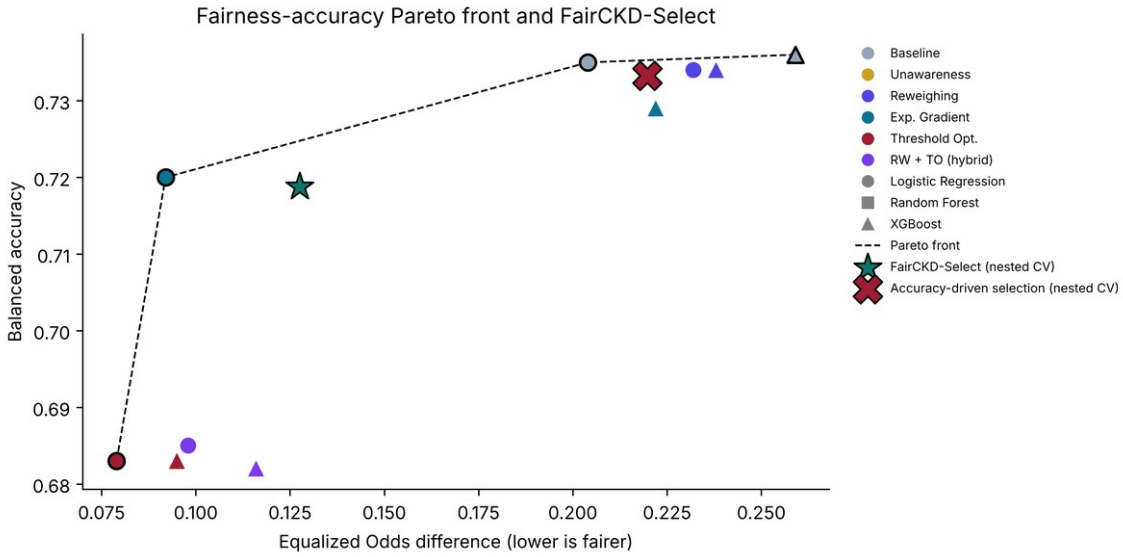
4b. FairCKD-Select (proposed method)

FairCKD-Select chooses, inside each training fold, the classifier and mitigation method with the smallest Equalized Odds gap among those whose balanced accuracy is within 3 points of the best unmitigated model (nested cross-validation). It is compared with conventional accuracy-driven selection.

Strategy	Accuracy	Balanced acc.	Recall	EOpp diff	EOdds diff	DP diff
FairCKD-Select	0.750 ± 0.001	0.719 ± 0.004	0.669 ± 0.012	0.128 ± 0.056	0.128 ± 0.056	0.083 ± 0.012
Accuracy-driven selection	0.755 ± 0.000	0.733 ± 0.002	0.698 ± 0.005	0.220 ± 0.022	0.220 ± 0.022	0.120 ± 0.003

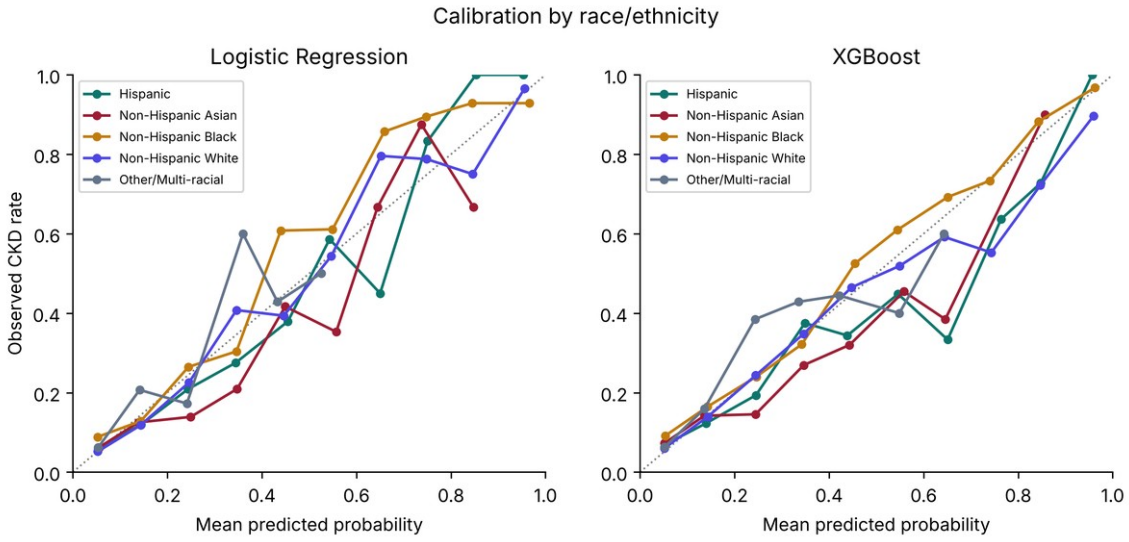
Pipelines chosen by FairCKD-Select across outer folds

Model	Method	Times selected
Logistic Regression	Exp. Gradient	8
XGBoost	Exp. Gradient	2



4c. Calibration by group

Group	Patients	Observed CKD (%)	Mean predicted (%)	Calibration-in-the-large (pp)	ECE	Brier	Model
Hispanic	1204	15.116	16.278	1.162	0.025	0.101	Logistic Regression
Non-Hispanic Asian	709	13.822	16.345	2.523	0.036	0.101	Logistic Regression
Non-Hispanic Black	1121	22.391	19.598	-2.793	0.042	0.126	Logistic Regression
Non-Hispanic White	1762	20.488	20.753	0.265	0.022	0.115	Logistic Regression
Other/Multi-racial	267	17.603	15.418	-2.185	0.042	0.114	Logistic Regression
Hispanic	1204	15.116	15.873	0.757	0.031	0.103	XGBoost
Non-Hispanic Asian	709	13.822	14.548	0.726	0.038	0.103	XGBoost
Non-Hispanic Black	1121	22.391	19.528	-2.863	0.032	0.125	XGBoost
Non-Hispanic White	1762	20.488	21.171	0.683	0.017	0.117	XGBoost
Other/Multi-racial	267	17.603	14.773	-2.83	0.036	0.114	XGBoost

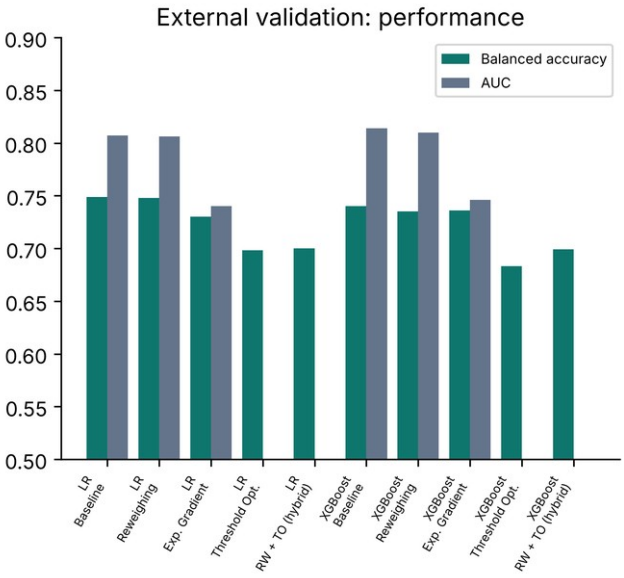


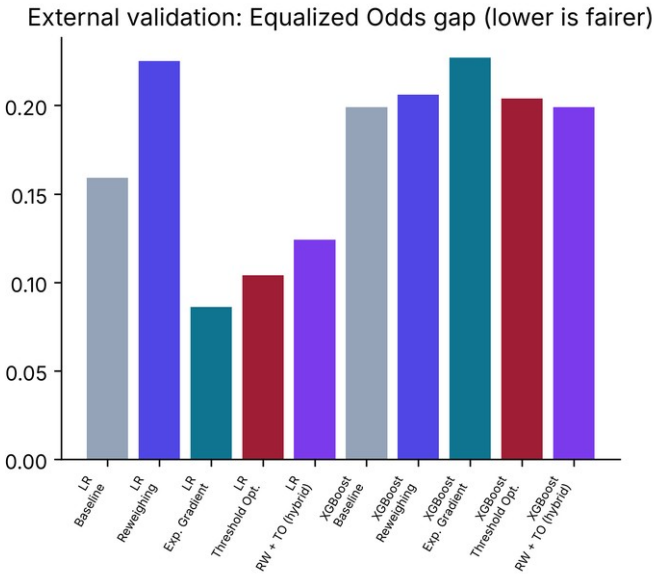
4d. Population-weighted estimates (NHANES survey weights)

W Accuracy	W Balanced acc.	W EOpp diff	W EOdds diff	W DP diff	W TPR [Hispanic]	W TPR [Non-Hispanic Asian]	W TPR [Non-Hispanic Black]	W TPR [Non-Hispanic White]	W
.787	0.731	0.275	0.275	0.106	0.498	0.549	0.636	0.719	0.
.784	0.694	0.192	0.192	0.098	0.459	0.470	0.512	0.620	0.
.781	0.739	0.186	0.186	0.099	0.564	0.560	0.642	0.730	0.
.850	0.703	0.184	0.184	0.072	0.357	0.428	0.508	0.542	0.
.806	0.694	0.142	0.142	0.065	0.451	0.447	0.482	0.582	0.
.848	0.715	0.142	0.142	0.084	0.459	0.432	0.528	0.564	0.
.809	0.722	0.174	0.174	0.087	0.517	0.476	0.580	0.647	0.

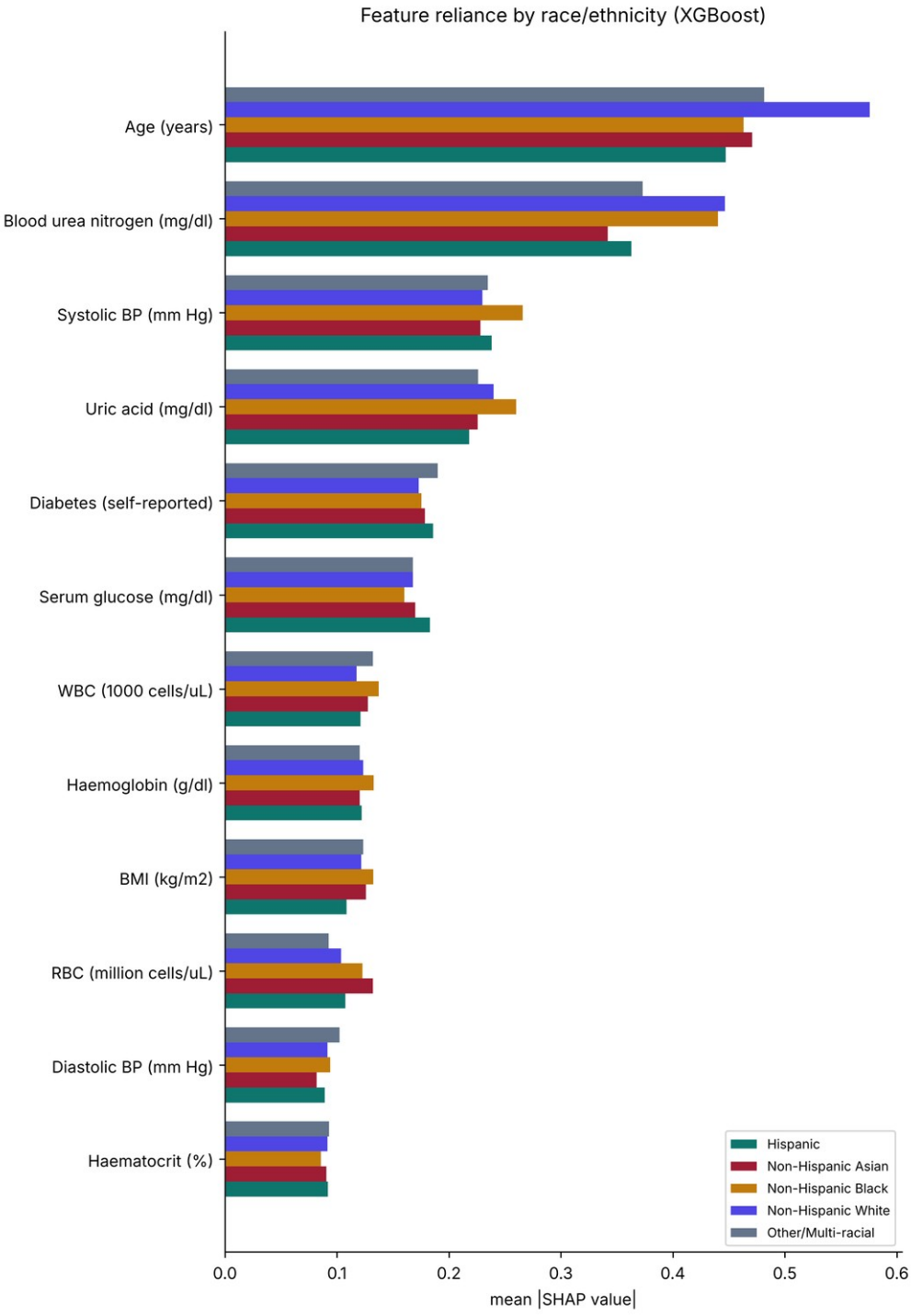
4e. External validation (trained on NHANES 2017–18, tested on NHANES 2015–16)

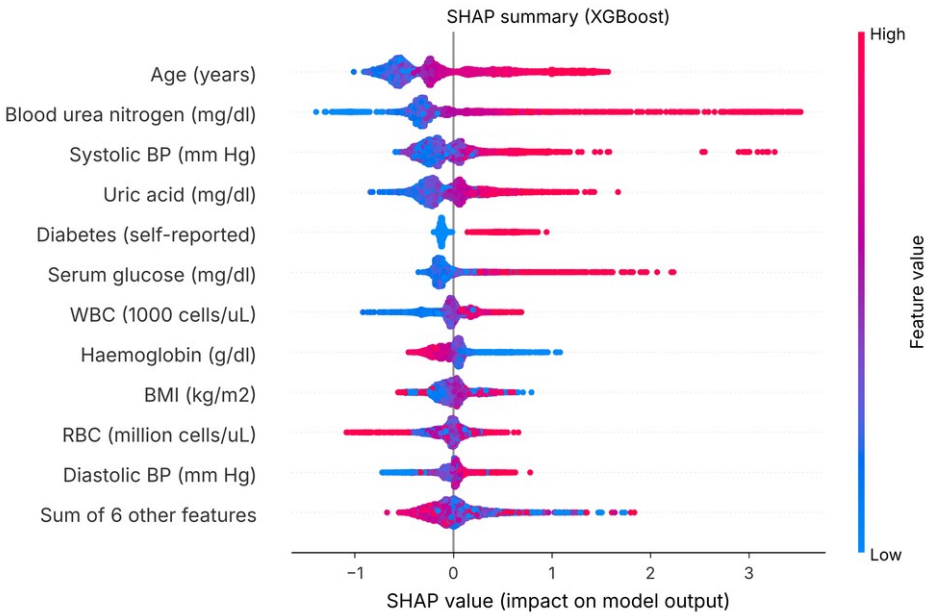
Model	Method	Accuracy	Balanced acc.	Recall	AUC	EOpp diff	EOdds diff
Logistic Regression	Baseline	0.784	0.749	0.696	0.807	0.159	0.159
Logistic Regression	Reweighing	0.785	0.748	0.694	0.806	0.225	0.225
Logistic Regression	Exp. Gradient	0.778	0.730	0.660	0.740	0.086	0.086
Logistic Regression	Threshold Opt.	0.726	0.698	0.656		0.104	0.104
Logistic Regression	RW + TO (hybrid)	0.725	0.700	0.663		0.124	0.124
XGBoost	Baseline	0.791	0.740	0.666	0.814	0.199	0.199
XGBoost	Reweighing	0.786	0.735	0.660	0.810	0.206	0.206
XGBoost	Exp. Gradient	0.809	0.736	0.627	0.746	0.227	0.227
XGBoost	Threshold Opt.	0.602	0.683	0.803		0.204	0.204
XGBoost	RW + TO (hybrid)	0.648	0.699	0.775		0.199	0.199





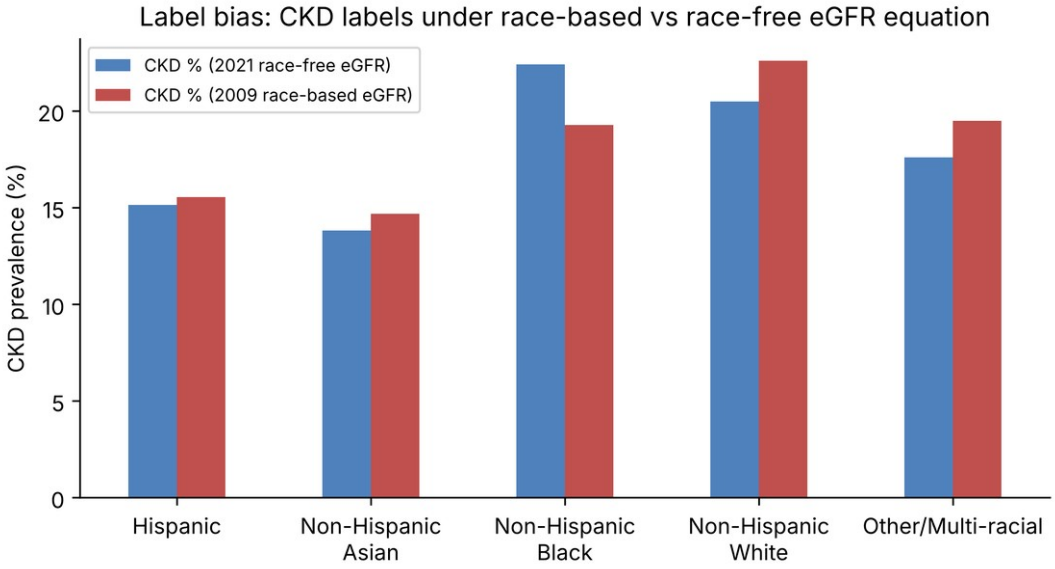
6. Explainability (SHAP)





7. Label bias (eGFR equation)

race	Adults	CKD % 2021 (race-free)	CKD % 2009 (race-based)	Change (pp)	Labels changed
Hispanic	1204	15.12	15.53	0.420	5
Non-Hispanic Asian	709	13.82	14.67	0.850	6
Non-Hispanic Black	1121	22.39	19.27	-3.12	35
Non-Hispanic White	1762	20.49	22.59	2.1	37
Other/Multi-racial	267	17.6	19.48	1.87	5



Protocol: repeated stratified 5-fold cross-validation (stratified on label × group); preprocessing and mitigation fitted on training folds only; mean ± SD over repeats. This system is a research prototype and not a medical device.